

NFC Book III Source Universality L4 Shell (Final Spine Book)

Schema v0.42.0

L4 Spine Release Candidate

Seven-Book Spine Complete

Generated by NFC Governance Engine v42

Contents

Book III L4 Shell (Final Spine Book)

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Book III is the final spine book. With this pilot, the full seven-book NFC spine has machine-readable L4 records. The Unified Coercive-Transport Inequality $C_{n+1} \leq \Theta_n C_n + E_n + B_n$ is the branch-agnostic load-bearing core [U]. The ACA Coercivity Chain [C] feeds Book VI's NS-4B Sobolev surrogate. This is an L4 spine-pilot shell — the canonical Book III is the authoritative source.

Book III Source Universality — L4 Generated Shell

This is the final spine book. With Book III, the full seven-book spine has machine-readable L4 records.

Book III's load-bearing core is unconditional: the Unified Coercive-Transport Inequality $C_{n+1} \leq \Theta_n C_n + E_n + B_n$ depends only on Book I's defect ledger and Book II's LS-2 boundary law. The ACA Coercivity Chain [C] feeds Book VI's NS-4B Sobolev surrogate.

Unconditional [U] Layer

Machine ID	St.	Canonical label
thm:III.depth-sum	[U]	Depth-Sum Convergence Theorem [U]
thm:III.unified-coercive	[U]	Unified Coercive-Transport Inequality [U] — three-source
thm:III.balance-composable	[U]	Balance Composability Theorem [U]
lem:III.charge-conservation	[U]	Charge Conservation Lemma [U]
thm:III.governing	[U]	Observational Transfer Theorem [U] — Book III Governing

ACA Chain [C] — Feeding Book VI NS-4B Sobolev Surrogate

Machine ID	St.	Canonical label
lem:III.cheeger-from-BUO	[C]	Cheeger Constant from BUO [C]
thm:III.ACA-chain	[C]	ACA Coercivity Chain [C] — ultracontractivity + Nash +

Machine ID	St.	Canonical label
thm:III.coercive-inequality	[C]	Finite Transfer Coercive Inequality [C] — conditional o

YM/NS Branch ACA Chain (now machine-traceable)

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thm:YM.7 [C] (YM endpoint - mass gap)
→ thm:VI.NS-4B-sobolev [C] (Sobolev surrogate - Book VI)
→ thm:III.ACA-chain [C] (ACA coercivity chain - Book III)
→ lem:III.cheeger-from-BU0 [C] (Cheeger from BU0 - Book III)
→ def:III.BU0 [D] (BU0 condition - Book III)
→ def:II.collar [D] (collar structure - Book II)

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Complete Seven-Book Spine (v42)

Book	Title	Character	Claims
I	Unconditional Source Architecture	All [U]	22
II	Locality and Boundary Structure	[U]+[C]	21
III	Source Universality/Depth Structure	[U]+[C]	17
IV	Universal Source and POT Category	Primarily [U]	19
V	Branch Legitimacy	Primarily [U]	19
VI	Continuum Bridge	Mostly [C]	18
VII	Governance Book	All [U]	19